Tyler Wong

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Summary

Computer engineer (B.S., University of Maryland) with hands-on experience in embedded firmware (STM32, FreeRTOS), hardware security, and IoT test/validation. Seeking embedded systems, firmware test, validation, or integration engineering roles in the Maryland/DC area.

Education

University of Maryland, College Park GPA: 3.395 College Park, MD
B.S. Computer Engineering, Minor in Technology Entrepreneurship May 2026
Relevant Coursework: Embedded Systems, Reverse Engineering & Hardware Security, Computer Architecture, Digital Logic Design, Cryptography, Programming Languages

Skills

Embedded & Firmware: C/C++, STM32 HAL, FreeRTOS, STM32CubeIDE, I2C, UART, MQTT, Arduino, x64/MIPS Assembly
Hardware & Testing: STM32L475 (ARM Cortex MCU), Raspberry Pi, Oscilloscope, Waveform Gen., FPGA, Soldering, Wireshark, Pspice, Vivado, Verilog, Fusion360
Software Dev. & Security: Ghidra (SRE), Python, Git, UNIX/WSL, Node-RED, Java, MATLAB, MySQL, MS Excel, XML, LaTeX

Experience

Untethered Labs (GateKeeper Proximity) — Cyber Security Operations Associate May 2023 – Present

- Provide technical support for a proximity-based hardware/software security product, manage inventory and order fulfillment, lead business projects, and maintain the company website.
- Built Python-based internal tools, including a PDF quote generator with a GUI that reads from an Excel pricing workbook.
- Researched and authored 30+ articles for an IT Wiki Database and created technical resources for sales, marketing, and technical support teams.

National Institute of Standards and Technology — Research Fellow, IoT Interoperability Testbench May 2025 – Aug. 2025

- Developed IoT testbenches simulating smart home integration over MQTT; physical (Raspberry Pi) and digital (Node-RED) testbench representations were tested for interoperability.
- Documented testbed development and usage, co-authored a publication for IEEE MNC 2026 [doi.org/rtx2], and delivered a lecture on methodology and results.

National Institute of Standards and Technology — Research Fellow, C-V2X Conformance Tool Jun. 2024 – Aug. 2024

- Developed an IoT Device Interoperability Analyzer Tool that assesses packet data for quantitative and structural conformance to standardized C-V2X protocols.
- Contributed to an official manuscript [doi.org/rtxz] documenting the tool's description, functionality, and usage.

Projects

Embedded Face Detector System — Embedded Systems Design Team Project Mar. 2026 – May 2026

- Built a Viola-Jones face detector in C++ as a Welter-C++ dataflow graph: a cascade of N boosted strong-classifier actors passing pointer-based image sub-window tokens, with no OpenCV.
- Wrote an AdaBoost training module that generates per-classifier configuration files, and Bash scripts to automate design space exploration of latency, detection rate, and false-positive rate on cross-compiled builds.

Pintos Operating System Kernel — File Systems and Virtual Memory Labs Jan. 2026 – Apr. 2026

- Implemented indexed inodes, subdirectories, a buffer cache with write-behind and read-ahead, demand-paged virtual memory with swap and mmap, and synchronization in a teaching kernel; passed the full 159-test suite. Debugged eviction races, teardown ordering, and kernel stack overflows. Mostly C++.

Password Strength Checker — Computer Systems Security Research Project Sep. 2025 – Dec. 2025

- Developed a program that classifies password strength using ML models trained on popular password datasets, plus a tool that turns memorable strings into strong passwords; co-authored the report on findings and methodology.

BOZOPAD — DIY 12-Key Macropad Kit (Club/Passion Project) Oct. 2021 – Jun. 2022

- Managed a limited run of the kit; co-developed the through-hole PCB, firmware (.hex) files, assembly instructions, interactive presentations, and organized fundraisers.